CAS SciFinder

Task History

Initiating Search

November 10, 2025, 2:03 PM

Search:

CAS Lexicon Search:

Birch reduction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (700)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (938) Filtered By: Document Type: **Journal**	**Reactions**	View Results

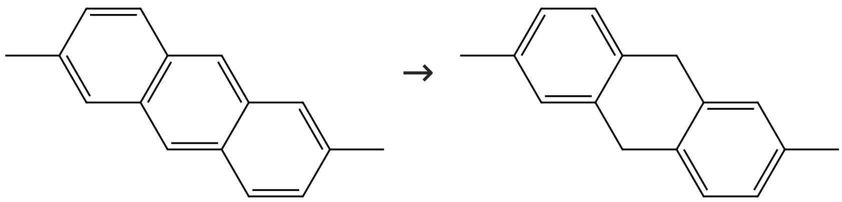
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (17) Suppliers (2)

31-243-CAS-8353636

Steps: **1** Yield: **100%**

1.1 **Reagents:** Lithium
Solvents: Tetrahydrofuran, Ammonia

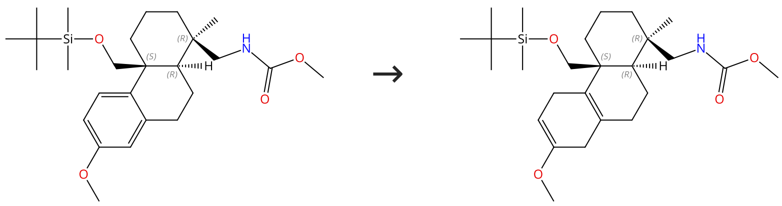
A diepoxide belonging to the rare C_i point group

By: Graham, Ron J.; et al

Journal of the Chinese Chemical Society (Taipei, Taiwan) (1992), 39(4), 279-83.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Relative stereochemistry shown Relative stereochemistry shown

31-243-CAS-7274492

Steps: **1** Yield: **100%**

1.1 **Reagents:** *tert*-Butanol, Lithium, Ammonia
Solvents: Ethanol, Tetrahydrofuran; 3 h, -40 °C

Experimental Protocols

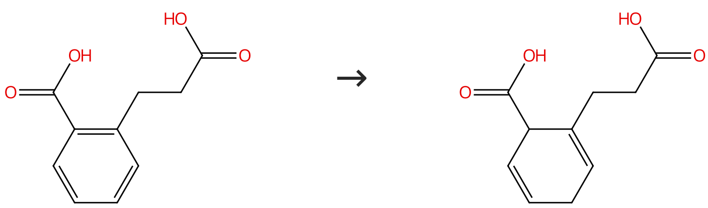
Studies toward the Total Synthesis of Nominine

By: Hutt, Oliver E.; et al

Journal of Organic Chemistry (2007), 72(26), 10130-10140.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (63)

31-243-CAS-239752

Steps: **1** Yield: **100%**

1.1 **Reagents:** Ammonia
Solvents: Tetrahydrofuran

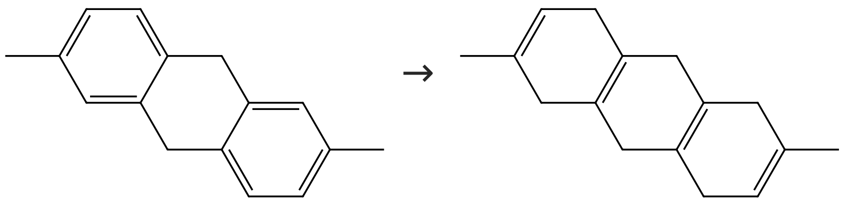
Synthesis based on cyclohexadienes. Part-27. Synthesis of functionalized tricyclo[5.2.2.$0^{1,5}$]undecanes

By: Rao, G. S. R. Subba; et al

Journal of the Indian Chemical Society (1997), 74(11-12), 961-969.

Scheme 4 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (2)

Suppliers (2)

31-243-CAS-10494795

Steps: **1** Yield: **100%**

1.1 **Reagents:** Lithium
Solvents: Ethanol, Methylamine

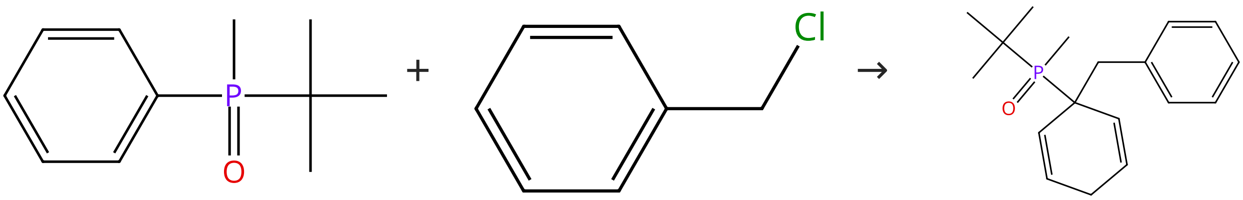
A diepoxide belonging to the rare C_i point group

By: Graham, Ron J.; et al

Journal of the Chinese Chemical Society (Taipei, Taiwan) (1992), 39(4), 279-83.

Scheme 5 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (3)

Suppliers (65)

Supplier (1)

31-243-CAS-9946106

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium, Ammonia; 15 min, -78 °C
1.2 5 min, -78 °C
1.3 15 min, -78 °C
1.4 **Reagents:** Ammonium chloride; -78 °C

Experimental Protocols

In Situ Dearomatisation/Alkylation of Arylphosphane Deriva tives

By: Stankevic, Marek; et al

European Journal of Organic Chemistry (2012), 2012(13), 2521-2534, S2521/1-S2521/64.

Scheme 6 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (19)

Suppliers (87)

31-243-CAS-815565

Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium, Ammonia; 15 min, -78 °C
1.2 5 min, -78 °C
1.3 15 min, -78 °C
1.4 **Reagents:** Ammonium chloride; -78 °C

Experimental Protocols

In Situ Dearomatisation/Alkylation of Arylphosphane Deriva tives

By: Stankevic, Marek; et al

European Journal of Organic Chemistry (2012), 2012(13), 2521-2534, S2521/1-S2521/64.

Scheme 7 (1 Reaction)

Steps: **1** Yield: **100%**

Suppliers (85) Suppliers (13)

31-243-CAS-14861925	Steps: **1** Yield: **100%**
1.1	**Reagents:** Water, Ammonium chloride **Solvents:** Water

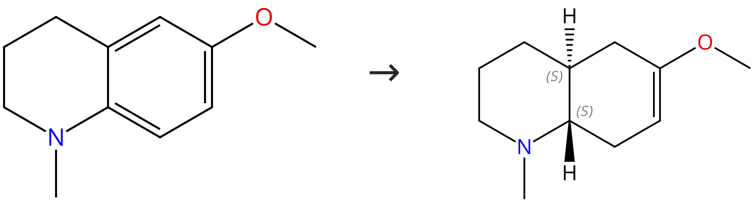
The metal-ammonia reduction of 1-naphthoic acid

By: Rabideau, Peter W.; et al

Synthetic Communications (1980), 10(8), 627-32.

Scheme 8 (1 Reaction)

Steps: **1** Yield: **100%**



Relative stereochemistry shown

Suppliers (11)

31-243-CAS-2562719	Steps: **1** Yield: **100%**
1.1	**Reagents:** Lithium, Ammonia **Solvents:** Tetrahydrofuran; reflux; 1 h, reflux
1.2	**Reagents:** Ethanol
1.3	**Reagents:** Water **Solvents:** Dichloromethane

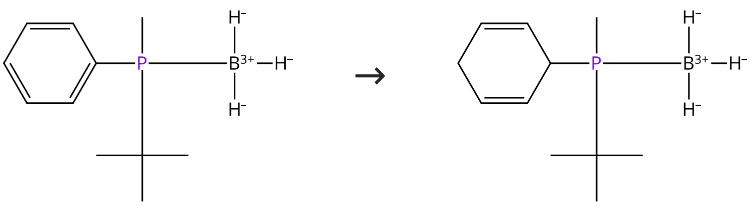
Synthesis of 9-azasteroid partial structures via Birch reduction as key step

By: Stanetty, Peter; et al

ARKIVOC (Gainesville, FL, United States) (2005), (5), 83-95.

Scheme 9 (1 Reaction)

Steps: **1** Yield: **100%**



31-243-CAS-10962269	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium **Solvents:** Tetrahydrofuran, Ammonia; 5 min, - 70 °C
1.2	**Reagents:** Ammonium chloride

Experimental Protocols

Birch reduction of aryldialkylphosphine-boranes

By: Stankevic, Marek; et al

Tetrahedron Letters (2009), 50(50), 7093-7095.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**

Suppliers (76)
Suppliers (77)

31-243-CAS-21417742 Steps: **1** Yield: **100%**

1.1 **Reagents:** Lithium
Solvents: Diethyl ether; 15 min, 0 °C

1.2 **Reagents:** Ammonia
Solvents: *tert*-Butanol; 0 °C; 7 h, 0 °C; overnight, rt

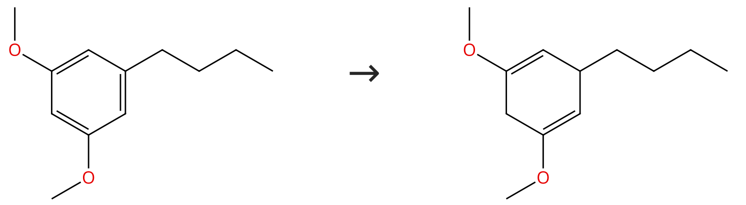
Experimental Protocols

Bromination of quinoxaline and derivatives: Effective synthesis of some new brominated quinoxalines

By: Ucar, Sefa; et al

Tetrahedron (2017), 73(12), 1618-1632.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (11)

31-243-CAS-15468612 Steps: **1** Yield: **100%**

1.1 **Reagents:** *tert*-Butanol, Ammonia
Solvents: Tetrahydrofuran; rt

1.2 **Reagents:** Lithium; -33 °C; -33 °C → rt

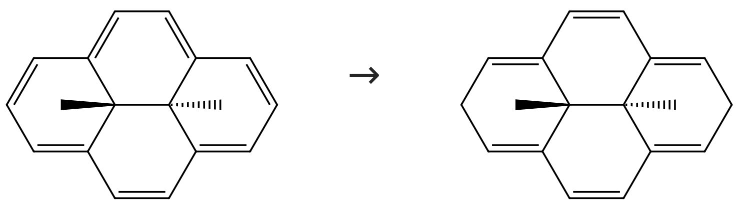
Experimental Protocols

Synthesis of chiloglottones: Semiochemicals from sexually deceptive orchids and their pollinators

By: Poldy, Jacqueline; et al

Organic & Biomolecular Chemistry (2009), 7(20), 4296-4300.

Scheme 12 (1 Reaction) Steps: **1** Yield: **99%**



Relative stereochemistry shown
Relative stereochemistry shown

Suppliers (3)
Supplier (1)

31-243-CAS-7326077 Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium hydroxide
Solvents: Methanol, Tetrahydrofuran; 5 min, rt

1.2 **Reagents:** Hydrochloric acid, Titanium chloride ($TiCl_3$)
Solvents: Water; 30 min

A novel Birch reduction of aromatic compounds using aqueous titanium trichloride: anions of trans-10b,10c-dimethyl-2,7,10b,10c-tetrahydropyrene

By: Jiang, Jianping; et al

Tetrahedron Letters (2003), 44(6), 1271-1274.

Scheme 13 (1 Reaction) Steps: **1** Yield: **99%**

31-243-CAS-21277891 Steps: **1** Yield: **99%**

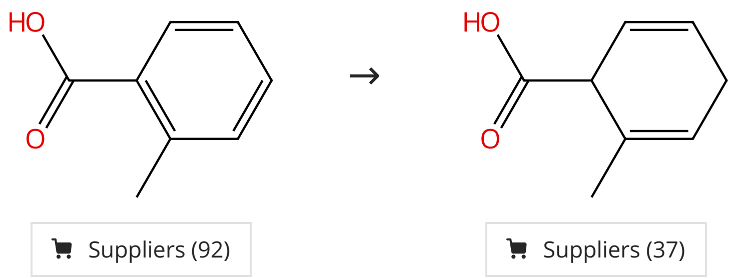
1.1 **Reagents:** Lithium, Ammonia; 12 h, -78 °C

Synthesis of selectively fluorinated cyclohexanes: The observation of phenonium rearrangements during deoxyfluorination reactions on cyclohexane rings with a vicinal phenyl substituent

By: Bykova, Tetiana; et al

Journal of Fluorine Chemistry (2015), 179, 188-192.

Scheme 14 (2 Reactions) Steps: **1** Yield: **97-99%**



31-614-CAS-45308375 Steps: **1** Yield: **99%**

1.1 **Reagents:** Ethylenediamine, Calcium
Solvents: Tetrahydrofuran; 15 min, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; pH 1 - 2, rt

Experimental Protocols

Highly efficient and air-tolerant calcium-based Birch reduction using mechanochemistry

By: Kubota, Koji; et al

Chemistry Letters (2024), 53(4), upae060.

31-614-CAS-24835596 Steps: **1** Yield: **97%**

1.1 **Catalysts:** Ethylenediamine
Solvents: Tetrahydrofuran; rt → 0 °C

1.2 **Reagents:** Lithium; 30 min, 0 °C

1.3 **Reagents:** Ammonium chloride
Solvents: Water

1.4 **Reagents:** Hydrochloric acid; pH 2, cooled

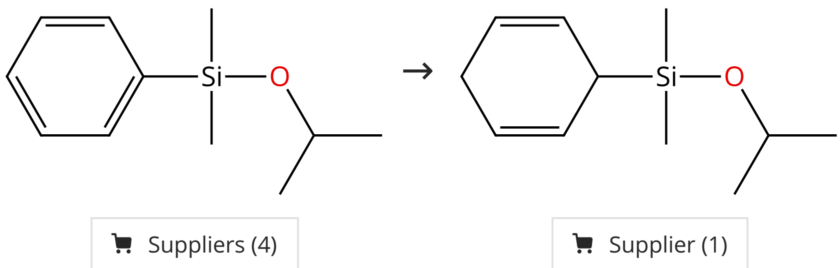
Experimental Protocols

Scalable Birch reduction with lithium and ethylene diamine in tetrahydrofuran

By: Burrows, James; et al

Science (Washington, DC, United States) (2021), 374(6568), 741-746.

Scheme 15 (1 Reaction) Steps: **1** Yield: **99%**



31-243-CAS-2367185 Steps: **1** Yield: **99%**

1.1 **Reagents:** 1-Butanol, Hexamethylphosphoramide, Aluminum, Lithium chloride
Solvents: Tetrahydrofuran; 5 h, rt; 3 h, rt

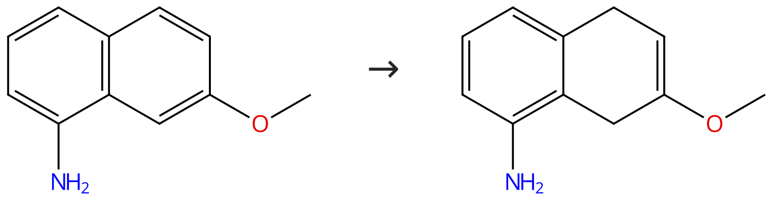
1.2 **Reagents:** Hydrochloric acid
Solvents: Pentane, Water; -78 °C

Desymmetrization of cyclohexa-1,4-dienes - a straight forward route to cyclic and acyclic polyhydroxylated systems

By: Landais, Yannick; et al

European Journal of Organic Chemistry (2002), (23), 4037-4053.

Scheme 16 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (61) Suppliers (2)

31-243-CAS-18320705 Steps: **1** Yield: **99%**

1.1 **Reagents:** Ammonia
Solvents: *tert*-Butanol, Tetrahydrofuran; rt → -78 °C

1.2 **Reagents:** Lithium; 1.5 h, -78 °C

1.3 **Solvents:** Methanol, Water; 1 h, -78 °C; -78 °C → rt

Experimental Protocols

The development of a short route to the API ropinirole hydrochloride

By: Yousuf, Zeshan; et al

Organic & Biomolecular Chemistry (2015), 13(42), 10532-10539.

Scheme 17 (1 Reaction) Steps: **1** Yield: **99%**

Suppliers (83) Suppliers (87) Suppliers (4)

31-243-CAS-21282070 Steps: **1** Yield: **99%**

1.1 **Reagents:** Lithium
Solvents: Ammonia; -60 °C; 30 min, -60 °C

1.2 5 min, -60 °C; overnight, -60 °C → rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; pH 1 - 2, rt

Experimental Protocols

Regioselective Arene Functionalization: Simple Substitution of Carboxylate by Alkyl Groups

By: Krueger, Tobias; et al

Chemistry - A European Journal (2009), 15(44), 12082-12091.

Scheme 18 (2 Reactions) Steps: **1** Yield: **98-99%**

Suppliers (92) Suppliers (87) Suppliers (3)

31-243-CAS-21374063 Steps: **1** Yield: **99%**

1.1 **Reagents:** Lithium, Ammonia
Solvents: Tetrahydrofuran; -78 °C; 15 min, - 78 °C
1.2 90 min, - 78 °C; overnight, - 78 °C → rt
1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

Synthesis of precursors of phomactins using [2, 3]-Wittig rearrangements

By: Shapland, Peter D. P.; et al

Tetrahedron (2009), 65(21), 4201-4211.

31-243-CAS-21281648 Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium
Solvents: Ammonia; -60 °C; 30 min, - 60 °C
1.2 5 min, - 60 °C; overnight, - 60 °C → rt
1.3 **Reagents:** Hydrochloric acid
Solvents: Water; pH 1 - 2, rt

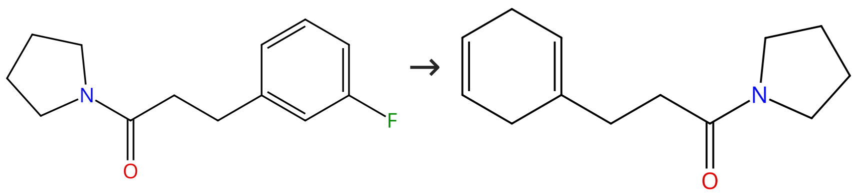
Experimental Protocols

Regioselective Arene Functionalization: Simple Substitution of Carboxylate by Alkyl Groups

By: Krueger, Tobias; et al

Chemistry - A European Journal (2009), 15(44), 12082-12091.

Scheme 19 (1 Reaction) Steps: **1** Yield: **99%**



Supplier (1)

31-243-CAS-18918198 Steps: **1** Yield: **99%**

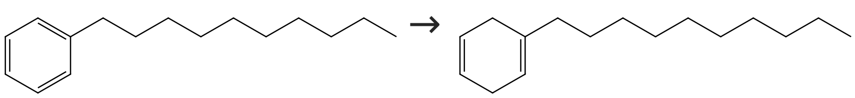
1.1 **Reagents:** Sodium, 15-Crown-5
Solvents: Tetrahydrofuran; 5 min, 0 °C
1.2 **Reagents:** Isopropanol
Solvents: Tetrahydrofuran; 30 min, 0 °C
1.3 **Reagents:** Sodium bicarbonate
Solvents: Water

A Practical and Chemoselective Ammonia-Free Birch Reduction

By: Lei, Peng; et al

Organic Letters (2018), 20(12), 3439-3442.

Scheme 20 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (54)

31-614-CAS-39662651 Steps: **1** Yield: **99%**

1.1 **Reagents:** Glucose, 1,3-Dimethyl-2-imidazolidinone, Sodium; 15 min, rt
1.2 **Reagents:** Methanol, Water

Experimental Protocols

Mechanochemistry enabling highly efficient Birch reduction using sodium lumps and D-(+)-glucose

By: Kondo, Keisuke; et al

Chemical Science (2024), 15(12), 4452-4457.

Scheme 21 (1 Reaction)

Steps: **1** Yield: **99%**

Suppliers (11)

Suppliers (3)

31-243-CAS-1427529

Steps: **1** Yield: **99%**

1.1 **Reagents:** Potassium, Ammonia
Solvents: *tert*-Butanol, Tetrahydrofuran

1.2 **Reagents:** Ammonium chloride

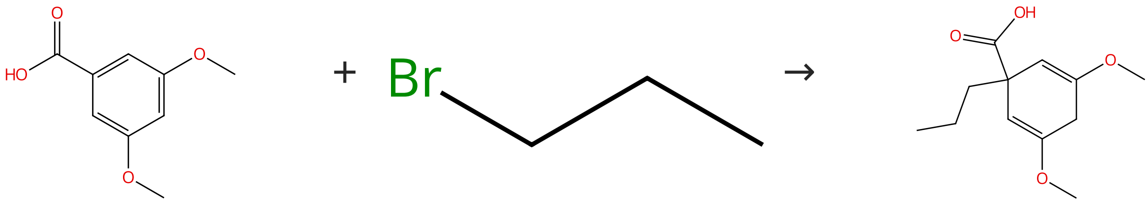
A convenient synthesis of 2-methyl-3-oxoheptane-1,7-dicarboxylic esters and amides

By: Elliott, Mark C.; et al

Tetrahedron Letters (2004), 45(14), 2899-2901.

Scheme 22 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (96)

Suppliers (62)

Suppliers (2)

31-066-CAS-4543472

Steps: **1** Yield: **99%**

1.1 **Reagents:** Ammonia
Solvents: Tetrahydrofuran; rt

1.2 **Reagents:** Lithium; -33 °C

1.3 -33 °C

1.4 **Reagents:** Hydrochloric acid
Solvents: Water; pH 3 - 4, 0 °C

Experimental Protocols

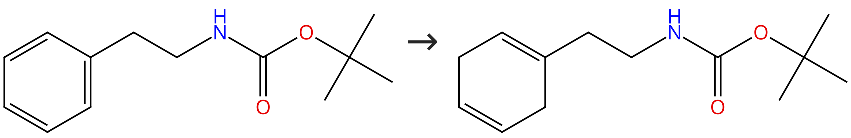
Synthesis of chiloglottones: Semiochemicals from sexually deceptive orchids and their pollinators

By: Poldy, Jacqueline; et al

Organic & Biomolecular Chemistry (2009), 7(20), 4296-4300.

Scheme 23 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (26)

31-614-CAS-31118674

Steps: **1** Yield: **99%**

1.1 **Reagents:** *tert*-Butanol, 1,3-Dimethyl-2-imidazolidinone, Sodium
Solvents: Tetrahydrofuran; 0 °C; 15 min, 0 °C

1.2 **Reagents:** Water; 0 °C

Experimental Protocols

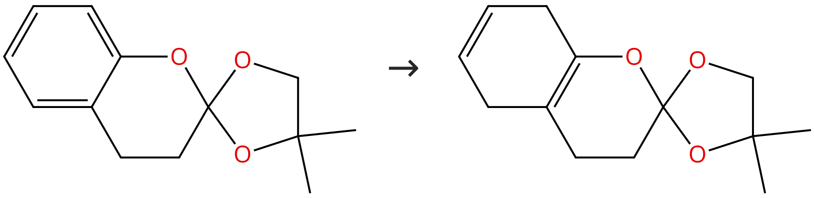
Birch Reduction of Arenes Using Sodium Dispersion and DMI under Mild Conditions

By: Asako, Sobi; et al

Chemistry Letters (2022), 51(1), 38-40.

Scheme 24 (1 Reaction)

Steps: **1** Yield: **99%**



Supplier (1)

Supplier (1)

31-243-CAS-12983540

Steps: **1** Yield: **99%**

1.1 **Reagents:** *tert*-Butanol, Lithium
Solvents: Tetrahydrofuran, Ammonia

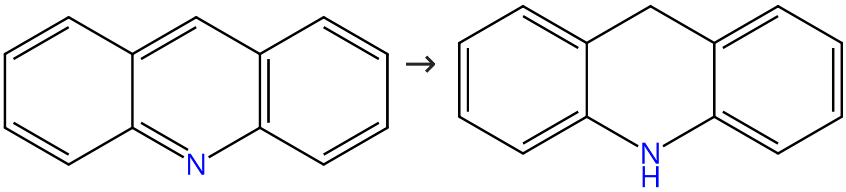
Enolic ortho esters. I. Preparation and Birch reduction of some coumarinoid ortho esters

By: Collins, David J.; et al

Australian Journal of Chemistry (1989), 42(8), 1235-48.

Scheme 25 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (74)

Suppliers (57)

31-243-CAS-21058122

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium, 15-Crown-5
Solvents: Tetrahydrofuran; 5 min, 0 °C

1.2 **Reagents:** Isopropanol
Solvents: Tetrahydrofuran; 5 min, 0 °C

1.3 **Reagents:** Sodium bicarbonate
Solvents: Water

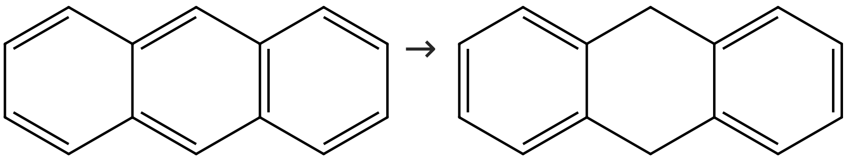
A Practical and Chemoselective Ammonia-Free Birch Reduction

By: Lei, Peng; et al

Organic Letters (2018), 20(12), 3439-3442.

Scheme 26 (4 Reactions)

Steps: **1** Yield: **97-99%**



Suppliers (114)

Suppliers (68)

31-243-CAS-8420802

Steps: **1** Yield: **99%**

1.1 **Reagents:** Sodium (stabilized in silica gel)
Solvents: 1,4-Dioxane; 15 min, 23 °C

1.2 **Reagents:** *tert*-Butanol; 5 min; 30 min, 29 °C; 2 h; cooled

1.3 **Solvents:** Methanol, Water; cooled; 2 h, rt

Experimental Protocols

Ammonia-free Birch reductions with sodium stabilized in silica gel, Na-SG(I)

By: Costanzo, Michael J.; et al

Tetrahedron Letters (2009), 50(39), 5463-5466.

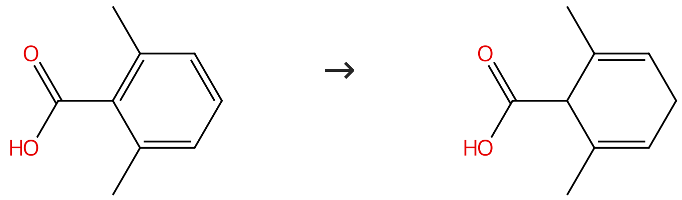
31-243-CAS-21342838	Steps: **1** Yield: **98%**	**Birch Reduction of Aromatic Compounds by Inorganic Electride $[Ca_2N]^{+\bullet}e^-$ in an Alcoholic Solvent: An Analogue of Solvated Electrons**
1.1	**Reagents:** Calcium nitride (Ca_2N), Electron **Solvents:** Isopropanol, Hexamethylphosphoramide; 24 h, 65 °C	By: Yoo, Byung Il; et al
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water	Journal of Organic Chemistry (2018), 83(22), 13847-13853.
Experimental Protocols		

31-243-CAS-21344007	Steps: **1** Yield: **98%**	**A Practical and Chemoselective Ammonia-Free Birch Reduction**
1.1	**Reagents:** Sodium, 15-Crown-5 **Solvents:** Tetrahydrofuran; 5 min, 0 °C	By: Lei, Peng; et al
1.2	**Reagents:** Isopropanol **Solvents:** Tetrahydrofuran; 5 min, 0 °C	Organic Letters (2018), 20(12), 3439-3442.
1.3	**Reagents:** Sodium bicarbonate **Solvents:** Water	

31-243-CAS-5319494	Steps: **1** Yield: **97%**	**Birch reductions at room temper ature with alkali metals in silica gel (Na_2K-SG(I))**
1.1	**Reagents:** Potassium alloy, nonbase, K,Na (in Silica Gel, stage I) **Solvents:** Tetrahydrofuran; 6 - 8 h, rt	By: Nandi, Partha; et al
1.2	**Reagents:** Water	Journal of Organic Chemistry (2009), 74(16), 5790-5792.
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (88)

Supplier (1)

31-614-CAS-45308376	Steps: **1** Yield: **99%**	**Highly efficient and air-tolerant calcium-based Birch reduction using mechanochemistry**
1.1	**Reagents:** Ethylenediamine, Calcium **Solvents:** Tetrahydrofuran; 15 min, rt	By: Kubota, Koji; et al
1.2	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 1 - 2, rt	Chemistry Letters (2024), 53(4), upae060.
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: **1** Yield: **98%**

Suppliers (92)

Suppliers (23)

Suppliers (3)

31-243-CAS-21375433 Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium, Ammonia
Solvents: Ammonia

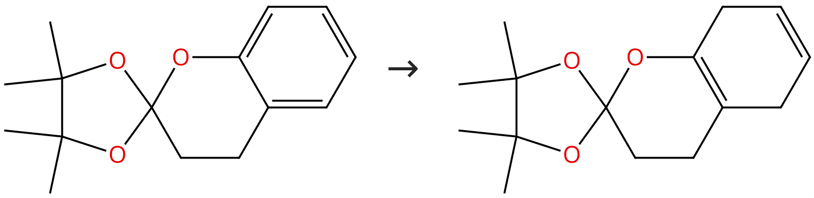
1.2 -

Substituent Effects in the Highly Regioselective and Diastereoselective Ene Reaction of Singlet Oxygen with Chiral Cyclohexadienes

By: Linker, Torsten; et al

Journal of the American Chemical Society (1995), 117(10), 2694-7.

Scheme 29 (1 Reaction) Steps: **1** Yield: **98%**



Supplier (1)

31-243-CAS-2002963 Steps: **1** Yield: **98%**

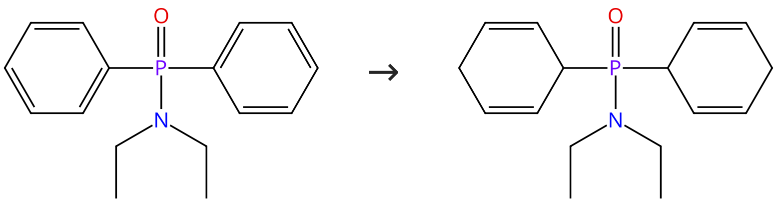
1.1 **Reagents:** *tert*-Butanol, Lithium
Solvents: Tetrahydrofuran, Ammonia

Enolic ortho esters. I. Preparation and Birch reduction of some coumarinoid ortho esters

By: Collins, David J.; et al

Australian Journal of Chemistry (1989), 42(8), 1235-48.

Scheme 30 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (6)

31-243-CAS-12401191 Steps: **1** Yield: **98%**

1.1 **Reagents:** Sodium, Ammonia; 15 min, -78 °C

1.2 **Solvents:** Tetrahydrofuran; 5 min, -78 °C

1.3 **Reagents:** Ammonium chloride

Experimental Protocols

The reactivity of arylphosphorus acid amides under Birch reduction conditions

By: Stankevic, Marek; et al

European Journal of Organic Chemistry (2013), 2013(20), 4351-4371.

Scheme 31 (1 Reaction) Steps: **1** Yield: **98%**

Absolute stereochemistry shown, Rotation (-)

Suppliers (87)

Absolute stereochemistry shown, Rotation (-)

Supplier (1)

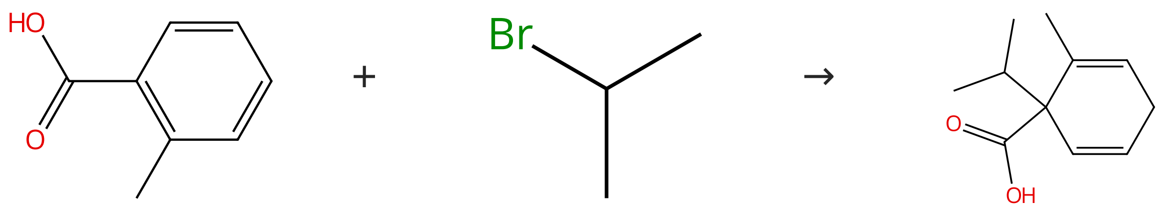
31-243-CAS-21026759	Steps: **1** Yield: **98%**
1.1	**Reagents:** Sodium **Solvents:** Tetrahydrofuran, Ammonia
1.2	-

Stereoselective reduction of chiral 2-furoic acid derivatives using group I metals in ammonia

By: Donohoe, Timothy J.; et al

Perkin 1 (2000), (22), 3724-3731.

Scheme 32 (2 Reactions) Steps: **1** Yield: **98%**



Suppliers (92) Suppliers (67) Suppliers (3)

31-066-CAS-1883748	Steps: **1** Yield: **98%**
1.1	**Reagents:** Lithium **Solvents:** Ammonia; -60 °C; 30 min, - 60 °C
1.2	5 min, -60 °C; overnight, -60 °C → rt
1.3	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 1 - 2, rt

Experimental Protocols

Regioselective Arene Functionalization: Simple Substitution of Carboxylate by Alkyl Groups

By: Krueger, Tobias; et al

Chemistry - A European Journal (2009), 15(44), 12082-12091.

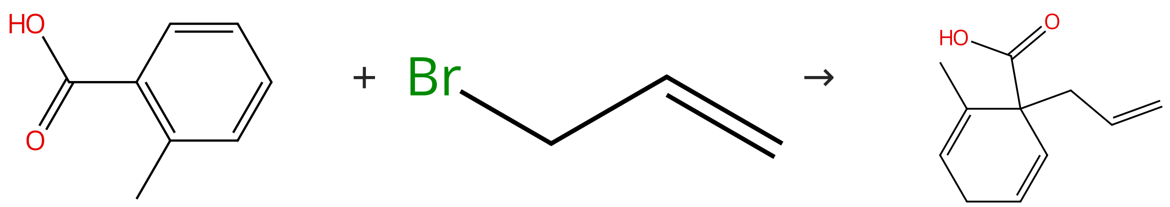
31-066-CAS-6445441	Steps: **1** Yield: **98%**
1.1	**Reagents:** Lithium, Ammonia **Solvents:** Ammonia
1.2	-

Substituent Effects in the Highly Regioselective and Diastereoselective Ene Reaction of Singlet Oxygen with Chiral Cyclohexadienes

By: Linker, Torsten; et al

Journal of the American Chemical Society (1995), 117(10), 2694-7.

Scheme 33 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (92) Suppliers (57) Suppliers (3)

31-066-CAS-4228407	Steps: **1** Yield: **98%**
1.1	**Reagents:** Lithium **Solvents:** Ammonia; -60 °C; 30 min, - 60 °C
1.2	5 min, -60 °C; overnight, -60 °C → rt
1.3	**Reagents:** Hydrochloric acid **Solvents:** Water; pH 1 - 2, rt

Experimental Protocols

Regioselective Arene Functionalization: Simple Substitution of Carboxylate by Alkyl Groups

By: Krueger, Tobias; et al

Chemistry - A European Journal (2009), 15(44), 12082-12091.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **98%**

Suppliers (91)

Suppliers (17)

31-243-CAS-13911199

Steps: **1** Yield: **98%**

1.1 **Reagents:** 1-Butanol, Hexamethylphosphoramide, Aluminum, Lithium chloride
Solvents: Tetrahydrofuran; 5 h, rt; 3 h, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Pentane, Water; -78 °C

Desymmetrization of cyclohexa-1,4-dienes - a straight forward route to cyclic and acyclic polyhydroxylated systems

By: Landais, Yannick; et al

European Journal of Organic Chemistry (2002), (23), 4037-4053.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **98%**

Suppliers (67)

Suppliers (5)

31-243-CAS-10236790

Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium, Ammonia, Ammonium chloride
Solvents: Isopropanol, Tetrahydrofuran

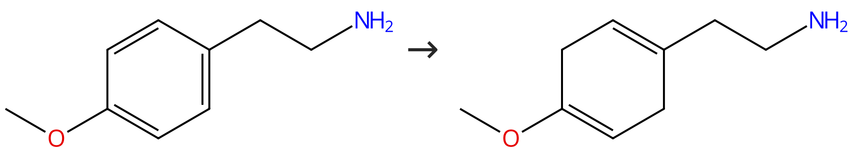
New synthesis of benzocyclobutene and bicyclo[4.2.0]octa-1(6),3-diene

By: Radlick, Phillip; et al

Journal of Organic Chemistry (1973), 38(19), 3412-3.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **98%**



Suppliers (59)

Suppliers (3)

31-243-CAS-1935883

Steps: **1** Yield: **98%**

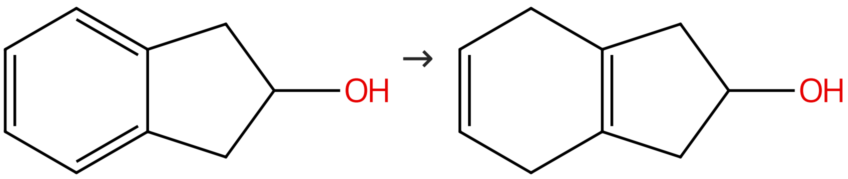
1.1 **Reagents:** Sodium, Ammonia
Solvents: Ethanol, Tetrahydrofuran; rt → -78 °C; 30 min, - 78 °C; 30 min, -78 °C

Stereoselective synthesis of (±)-γ-lycorane

By: Shao, Zhi-hui; et al

Hecheng Huaxue (2004), 12(2), 111-114.

Scheme 37 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (73)

Suppliers (2)

31-243-CAS-4578136 Steps: **1** Yield: **98%**

1.1 **Reagents:** Lithium, Ammonia
Solvents: Ethanol, Tetrahydrofuran; 1 h, -78 °C; 1 h, - 78 °C

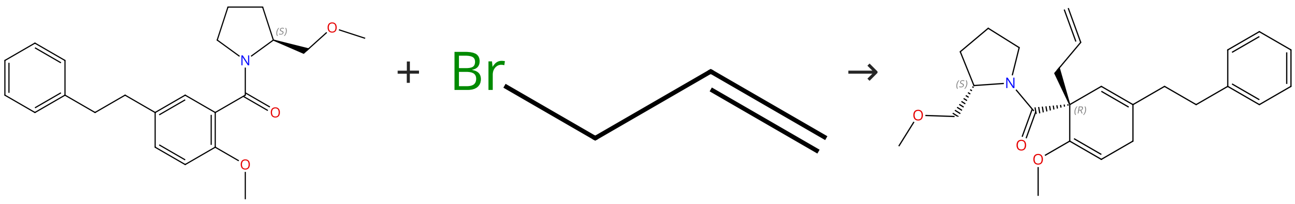
1.2 **Solvents:** Ethanol, Water; -78 °C; -78 °C → rt

Synthesis and characterization of 2,3a,5,6,7a-pentaacetoxy-octahydro-1H-indene from indan-2-ol

By: Balci, Neslihan; et al

Journal of Chemical Research (2009), (4), 248-251.

Scheme 38 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown

Suppliers (57)

Absolute stereochemistry shown

31-066-CAS-18314078 Steps: **1** Yield: **97%**

1.1 **Reagents:** *tert*-Butanol, Potassium, Ammonia
Solvents: Tetrahydrofuran; 20 min, -78 °C

1.2 **Reagents:** Isoprene; -78 °C

1.3 -78 °C → rt

Experimental Protocols

The enantioselective construction of tetracyclic diterpene skeletons with Friedel-Crafts alkylation and palladium-catalyzed cycloalkenylation reactions

By: Burke, Sarah J.; et al

Organic & Biomolecular Chemistry (2015), 13(9), 2726-2744.

Scheme 39 (1 Reaction) Steps: **1** Yield: **97%**

Suppliers (7)

Suppliers (87)

Suppliers (16)

31-243-CAS-21383180 Steps: **1** Yield: **97%**

1.1 **Reagents:** *tert*-Butanol, Lithium, Ammonia
Solvents: Tetrahydrofuran; -78 °C; 5 min, - 78 °C

1.2 -78 °C; 30 min, - 78 °C; - 78 °C → rt; overnight, rt

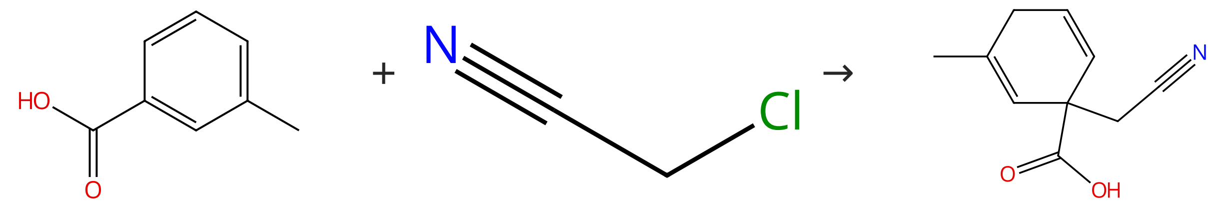
Experimental Protocols

Toward the Total Synthesis of Hygrocin B and Divergolide C: Construction of the Naphthoquinone-Azepinone Core

By: Nawrat, Christopher C.; et al

Organic Letters (2014), 16(7), 1896-1899.

Scheme 40 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (83) Suppliers (38) Supplier (1)

31-066-CAS-21178236 Steps: **1** Yield: **97%**

1.1 **Reagents:** Lithium, Ammonia; -78 °C; 1 h, -78 °C
1.2 2 min, -78 °C; overnight, rt
1.3 **Reagents:** Hydrochloric acid
Solvents: Water; pH 2

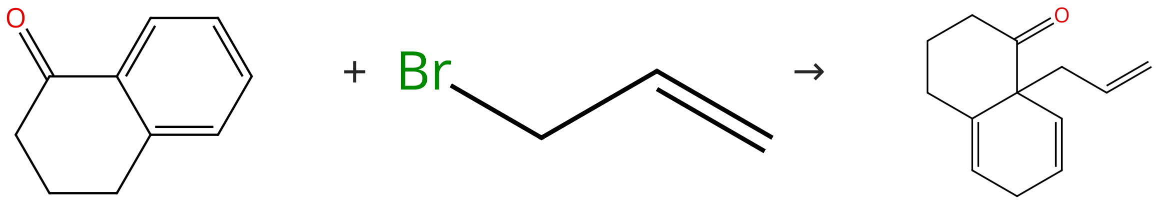
Experimental Protocols

Simple Synthesis of γ-Spirolactams by Birch Reduction of Benzoic Acids

By: Krueger, Tobias; et al

European Journal of Organic Chemistry (2017), 2017(6), 1074-1077.

Scheme 41 (1 Reaction) Steps: **1** Yield: **97%**



Suppliers (99) Suppliers (57)

31-066-CAS-4551103 Steps: **1** Yield: **97%**

1.1 **Reagents:** Ammonia
Solvents: Diethyl ether, *tert*-Butanol
1.2 **Reagents:** Potassium
1.3 **Reagents:** Lithium bromide
1.4 -
1.5 **Reagents:** Sodium chloride
Solvents: Diethyl ether
1.6 **Reagents:** Hydrochloric acid
1.7 **Solvents:** Diethyl ether

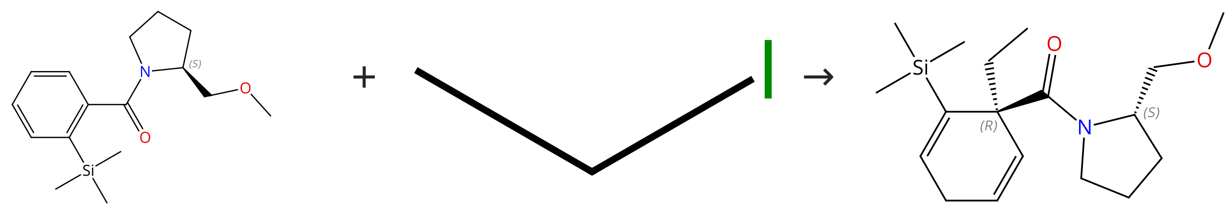
Experimental Protocols

Studies of Birch reductive alkylation of substi tuted α-tetralones

By: Labadie, Guillermo R.; et al

Synthetic Communications (2000), 30(22), 4065-4079.

Scheme 42 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (-)

Suppliers (66)

Absolute stereochemistry shown, Rotation (-)

Supplier (1)

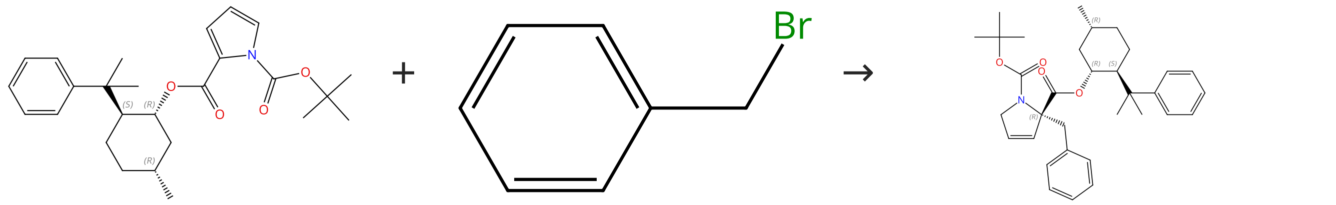
31-066-CAS-6446655	Steps: **1** Yield: **97%**
1.1	**Reagents:** *tert*-Butanol, Potassium **Solvents:** Tetrahydrofuran, Ammonia
1.2	**Reagents:** Lithium bromide **Solvents:** Tetrahydrofuran, Ammonia
1.3	**Reagents:** 1,3-Pentadiene **Solvents:** Tetrahydrofuran, Ammonia
1.4	**Solvents:** Tetrahydrofuran, Ammonia

Desymmetrization of Benzoic Acid in the Context of the Asymmetric Birch Reduction-Alkylation Protocol. Asymmetric Total Syntheses of (-)-Eburnamonine and (-)-Aspidosp ermidine

By: Schultz, Arthur G.; et al

Journal of Organic Chemistry (1997), 62(20), 6855-6861.

Scheme 43 (1 Reaction) Steps: **1** Yield: **96%**



Absolute stereochemistry shown

Suppliers (67)

Absolute stereochemistry shown

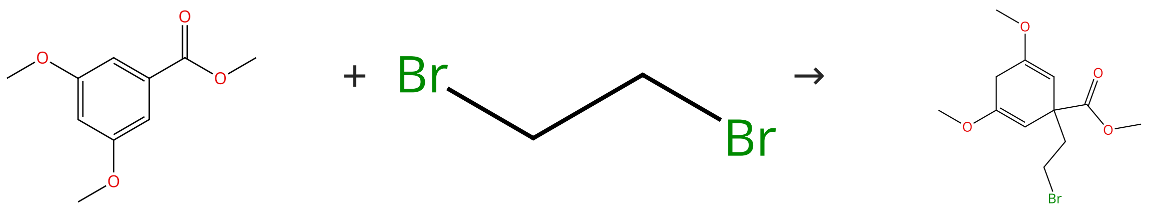
31-066-CAS-9958240	Steps: **1** Yield: **96%**
1.1	**Reagents:** Bis(2-methoxyethyl)amine, Lithium, Ammonia **Solvents:** Tetrahydrofuran
1.2	**Reagents:** Isoprene
1.3	-
1.4	**Reagents:** Ammonium chloride **Solvents:** Water

The stereoselective Birch reduction of pyrroles

By: Donohoe, Timothy J.; et al

Tetrahedron Letters (1999), 40(3), 435-438.

Scheme 44 (1 Reaction) Steps: **1** Yield: **96%**



Suppliers (76)

Suppliers (60)

Suppliers (2)

31-066-CAS-5666096	Steps: **1** Yield: **96%**
1.1	**Reagents:** *tert*-Butanol, Lithium, Ammonia **Solvents:** Tetrahydrofuran; -78 °C
1.2	-

Synthesis of an azaspirane via Birch reduction alkylation prompted by suggestions from a computer program

By: Tanaka, Akio; et al

Tetrahedron Letters (2006), 47(38), 6733-6737.